

Person	Time (in days)	Total work (in units)	Efficiency (in units/day)
A + B	15	60	$\frac{60}{15} = 4$
B + C	12	60	$\frac{60}{12} = 5$
C + A	10	60	$\frac{60}{10} = 6$

$$\Rightarrow 2 \times (A + B + C) = (4 + 5 + 6)$$

$$\Rightarrow (A + B + C) = 7.5$$

$$\begin{aligned} \text{Efficiency of B} &= (A + B + C) - (C + A) \\ &= (7.5) - (6) = 1.5 \end{aligned}$$

According to question,

(A + B + C) work for 6 days and completed (6 × 7.5) units work.

$$\begin{aligned} &= \text{Remaining work} = (60 - 45) \\ &= 15 \text{ units} \end{aligned}$$

Time taken by B to complete the remaining work = $15/1.5 = 10$ days

39. (d) The maximum marks for examination = x

According to question

$$(x \times 30\%) + 50 = (x \times 50\%) - 30$$

$$\text{Now, } x \times 30\% - x \times 50\% = -30 - 50$$

$$\Rightarrow 20x\% = 80 \Rightarrow x = 400$$

40. (d) Here, $M = 44$, $H = 9$

Therefore required angle,

$$\theta = \left(30H - \frac{11}{2}M^\circ \right)$$

$$= \left| 30 \times 9 - \left(\frac{11}{2} \right) \times 44 \right|^\circ$$

$$= |270 - 242|^\circ = 28^\circ$$

41. (d) CP for A = $360 \times \frac{100}{60} = ₹ 600$.

The price at which B sold to C = $600 + 29\%$ of $600 = 774$

The total cost price for B = $360 + 90 = ₹ 450$.

Profit for B = Selling price for B - Cost price for B = $774 - 450 = 324$

42. (d) Diameter of First Pizza = 14 cm; Radius = 7 cm

$$\text{Area of the first pizza} = \pi \times (7)^2 = 49\pi \text{ cm}^2.$$

Price for first pizza of $49\pi \text{ cm}^2 = ₹ 280$

Diameter of Second Pizza = 28 cm; Radius = 14 cm

$$\text{Area of the second pizza} = \pi \times (14)^2 = 196\pi \text{ cm}^2.$$

So, the of 2nd pizza

$$= \frac{280}{49\pi} \times 196\pi = ₹ 1120$$

So, the required discount = $1120 - 860 = 260$

43. (d) Subhman sold an article to Rinku for = ₹ 616

Cost price of the article for Subhman

$$= ₹ 616 \times \frac{100}{80} = ₹ 770$$

Final cost price of the article for Rinku

$$= 770 + 180 = ₹ 950$$

Selling price of the article for Rinku

$$= ₹ 770 + 28.56\% \text{ of } 770 = 770 + \frac{2}{7} \times 770 = 990$$

Profit of Rinku = $990 - 950 = 40$

Profit percentage on the article for Rinku = $\frac{40}{990} \times 100\% = 4.04\%$

44. (d) Area of the square = a^2

$$\Rightarrow a^2 = 22\sqrt{3}$$

$$\Rightarrow a = \sqrt{22\sqrt{3}}$$

$$\text{Diagonal of a square} = \sqrt{2} a$$

$$\text{Diagonal of the square} = \sqrt{2} \times \sqrt{22\sqrt{3}}$$

Side of the equilateral triangle

$$= \sqrt{2} \times \sqrt{22\sqrt{3}} \text{ [Diagonal of the square]}$$

Area of the equilateral triangle

$$= \left[\sqrt{\frac{3}{4}} \right] \left[\sqrt{2} \times (22\sqrt{3})^2 \right]$$

$$= \left[\sqrt{\frac{3}{4}} \right] \times 2 \times 22\sqrt{3} = 33$$

45. (b) $\frac{\sqrt[3]{13^3} + \sqrt[3]{14^3}}{3} = x\% \text{ of } 50$

$$\frac{13+14}{3} = x\% \text{ of } 50$$

$$\Rightarrow \frac{27}{3} = x\% \text{ of } 50$$

$$\Rightarrow 9 = \frac{x}{100} \times 50$$

$$\Rightarrow x = 18$$

46. (a) Given: The new speed is 80% of the usual speed

Total distance cover 2000 km in 52 hours

Take 2 hour's rest then return

Formula used:

$$\text{Average speed} = \frac{2ab}{a+b}$$

a = new speed, b = usual speed

Calculation:

$$\text{New speed} = \frac{80}{100} \text{ (usual speed)}$$

$$\Rightarrow \frac{\text{New speed}}{\text{Usual speed}} = \frac{4}{5}$$

New speed = $4u$, usual speed = $5u$

Time excluding the rest hour

$$= 52 - 2 = 50 \text{ h}$$

$$\text{Average speed} = \frac{2000}{50}$$

$$\text{Average speed} = 40 \text{ km/h}$$

$$\Rightarrow 2 \frac{4u \times 5u}{4u + 5u} = 40$$

$$\Rightarrow \frac{40u^2}{9u} = 40 \Rightarrow u = 9$$

$$\text{Since, } \text{km/h} \times \frac{5}{18} = \text{m/s}$$

New speed = $4u$

$$\text{New speed} = 9 \times 4 = 36 \times \frac{5}{18} = 10 \text{ m/s}$$

47. (c) Let the cost of the Horse be ₹ x
Then the cost of the Carriage will be ₹ $(30,000 - x)$

According to the question,

$$\Rightarrow 120\% \text{ of } x + 90\% \text{ of } (30,000 - x) = 102\% \text{ of } 30,000$$

$$\Rightarrow \frac{6}{5}x + \frac{9}{10}(30,000 - x) = 102 \times 300$$

$$\Rightarrow \frac{12x + 270000 - 9x}{10} = 30,600$$

$$\Rightarrow 3x + 2,70,000 = 3,06,000$$

$$\Rightarrow 3x = 3,06,000 - 2,70,000$$

$$\Rightarrow 3x = 36,000$$

$$\Rightarrow x = 12,000$$

$$\Rightarrow \text{The cost of Horse is ₹ } 12,000.$$

48. (b) Concept used:

LCM × HCF = Product of the two numbers

Calculation:

According to the question,

$$\text{HCF} = \frac{1}{20} \times \text{LCM} \quad \dots(i)$$

$$\text{LCM} - \text{HCF} = 456$$

$$\Rightarrow \text{LCM} - \left(\frac{1}{20} \times \text{LCM} \right) = 456$$

[from equation (i)]

$$\Rightarrow \frac{1}{20} \times \text{LCM} = 456$$

$$\Rightarrow \text{LCM} = \frac{456 \times 20}{19}$$

$$\Rightarrow \text{LCM} = 480$$

$$\text{Then HCF} = (480 - 456) = 24$$

Let's assume that the other number be k , then

$$k \times 96 = 24 \times 480$$

$$\Rightarrow k = \frac{11520}{96}$$

$$\Rightarrow k = 120$$

$$\Rightarrow \text{The correct answer is } 120$$

49. (d) Given: Average age of 12 students = 23 years

Average age of students including two teachers = 35 years

Age of science teacher

= (Age of mathematics teacher + 15) years

Formula:

Sum of total observation = Total number of observations $\times$ Average

Calculation:

Let, Age of mathematics teacher = x years

Age of science teacher = $(x + 15)$ years

Total age of 12 students

$\Rightarrow 23 \times 12 = 276$ years

Total age of students including two teachers

$\Rightarrow 35 \times 14 = 490$ years

According to the question,

$\Rightarrow 276 + x + (x + 15) = 490$

$\Rightarrow 276 + 2x + 15 = 490$

$\Rightarrow 2x + 291 = 490$

$\Rightarrow 2x = 490 - 291$

$\Rightarrow x = \frac{199}{2} = 99.5$ years

$\Rightarrow$ The age of mathematics teacher is 99.5 years.

50. (c) Calculation:

According to the question,

$$A = \frac{2}{5}B + B = \frac{7B}{5}$$

$$5A = 7B$$

$$10A = 14B \quad \dots(1)$$

Again,

$$B = \frac{3}{7}(A + C)$$

$$7B = 3A + 3C$$

$$5A = 3A + 3C \quad (\because 7B = 5A)$$

$$2A = 3C$$

$$10A = 15C \quad \dots(2)$$

From (1) & (2)

$$10A = 14B = 15C$$

$$A : B : C = \frac{1}{10} : \frac{1}{14} : \frac{1}{15}$$

Multiply with 210 (LCM of 10, 14 & 15)

$$A : B : C = \frac{210}{10} : \frac{210}{14} : \frac{210}{15}$$

$$A : B : C = 21 : 15 : 14$$

Therefore,

$$\text{Share of C} = 49000 \times \frac{14}{21+15+14}$$

$$= 49000 \times \frac{14}{50}$$

$$\Rightarrow \text{Share of C} = 13720/-$$

51. (a) The 1907 Indian National Congress Session at Surat was presided over by Ras Behari Ghosh. This session is historically significant because it marked the split between the Moderates and the Extremists within the Congress.

The Extremists, led by Bal Gangadhar Tilak, Lala Lajpat Rai, and Bipin Chandra Pal, advocated for immediate and direct action against British rule, including the use of boycotts and swadeshi (indigenous) goods.

In contrast, the Moderates, led by figures like Gopal Krishna Gokhale and Ras Behari Ghosh, pushed for gradual reform and more diplomatic approaches through petitions and discussions with the British authorities.

52. (c) The group of ministers during Shivaji Maharaj's time was called the Ashta Pradhan.

Ashta Pradhan translates to "Council of Eight" in English.

It was an administrative and advisory council. It is said that this council contributed to the military attacks on the Muslim Mughal Empire by Chhatrapati Shivaji Maharaj.

Shivaji Maharaj founded the Maratha Empire in India.

He found the harsh treatment and religious persecution of Hindus by Muslims completely unacceptable and unjust.

In 1674 AD, he was officially crowned as Chhatrapati at Raigarh.

53. (b) Gulbadan Begum wrote Humayun Nama, which is a biography of her brother, Emperor Humayun.

Abul Fazl is famous for writing the Ain-i-Akbari, a part of the Akbarnama which documents the administrative system of the Empire under Akbar.

Nizamuddin Ahmad authored the Tabaqat-i-Akbari, a detailed history of Akbar's reign.

Dara Shikoh translated the Upanishads into Persian, known as Sakinat-ul-Auliya, but this work is better known as Sirr-i-Akbar.

54. (b) The rock that is made up of molten magma is Igneous rock. Igneous rocks form from the solidification of molten magma.

If the magma cools and solidifies beneath the Earth's surface, it forms intrusive igneous rocks. If it cools and solidifies on the surface, it forms extrusive igneous rocks.

Granite, gabbro, basalt, are some of the examples of igneous rocks.

Intrusive igneous rocks form when magma cools and solidifies beneath the Earth's surface. Examples include diorite, gabbro, and granite.

Conversely, extrusive igneous rocks occur when magma crystallizes on the Earth's surface, resulting in smaller crystals. Common examples of these rocks are basalt, andesite, and dacite.

55. (d) The density of water is maximum at 4°C . At this temperature, water is at its densest and most compact form before it begins to expand slightly as it cools further towards freezing or heats up.

As water cools down to 4°C , it contracts like most substances, increasing in density. However, as it cools from 4°C to 0°C , it starts to expand.

This behaviour of water is called its anomalous Expansion of water.

This anomalous expansion is due to the hydrogen bonds between water molecules, which cause the molecules to arrange in a structured, less dense form as ice. This is why ice floats on liquid water.

56. (d) New Zealand announced the introduction of the Debbie Hockley Medal to honour the outstanding female cricketer.

The medal is named after Debbie Hockley, a legendary New Zealand cricketer who was the first woman to play 100 One Day Internationals and has made significant contributions to women's cricket both on and off the field.

57. (b) The first modern Olympic Games were indeed held in 1896 in Athens, Greece. This marked the revival of the Olympic Games, inspired by the ancient games held in Olympia, Greece.

The Olympic Games were not held during World War II due to the global turmoil. Specifically, the Games were cancelled in 1940 and 1944.

In 1940, Japan was originally selected to be the first Asian country to host the Olympics, which were supposed to be held in Tokyo. However, these Games were cancelled due to the outbreak of World War II.

58. (c) Women's cricket was introduced at the 2022 Commonwealth Games, starting on July 29 and ending with the final on August 7, 2022. These games took place in Birmingham, England.

The first match saw India competing against Australia.

The last time cricket was included in a multi-sport event was at the 1998 games in Kuala Lumpur.

The teams for the games were finalized by the end of January 2022.

The Commonwealth Games is an international multi-sport event featuring athletes from the Commonwealth of Nations. The first event took place in 1930. Except for 1942 and 1946, the games have been held every four years since their inception.